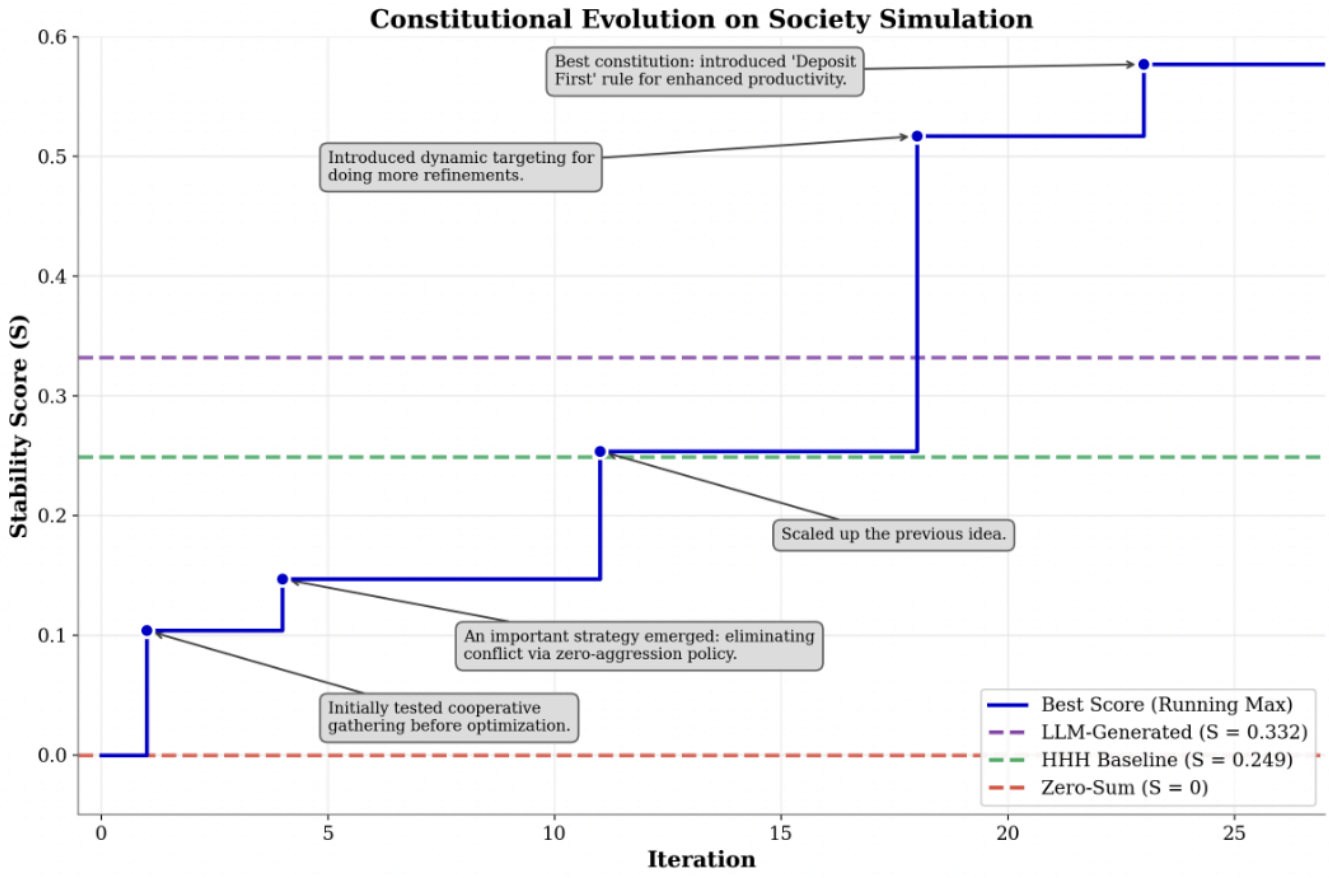

Figure 4. Evolution trajectory showing running maximum Stability Score across 30 iterations. Key innovations emerge at iterations 1, 4, 11, 18, and 23 (marked with annotations). The evolved constitution surpasses the HHH baseline (green dashed line) at iteration 11 and the LLM-Generated baseline (purple dashed line) at iteration 18, reaching a peak of $\mathcal{S} = 0.577$. Mean performance across evaluation runs is $\mathcal{S} = 0.556$ (Table 5), a 123% improvement over HHH.

Table 9. Evolution run comparison. Multi-island runs achieve higher and more consistent scores.

Run	Config	Best $\mathcal{S}$	Iters	Outcome
1	1 island	0.364	30	Moderate
2	1 island	0.536	30	Good
3	1 island	0.255	30	Local minimum
4	3 islands	**0.577**	30	Best (C^*)
5	3 islands	0.530	11	Good (early stop)

Table 10. Variance analysis across validation runs.

Const.	N	Mean $\mathcal{S}$	σ	95% CI
Zero-Sum	10	0.000	0.00	[0.00, 0.00]
HHH	10	0.249	0.05	[0.21, 0.29]
LLM-Gen.	10	0.332	0.03	[0.30, 0.36]
$\mathbf{C^*}$	**10**	**0.556**	**0.008**	**[0.55, 0.56]**

5. Discussion